

Agentic RAG



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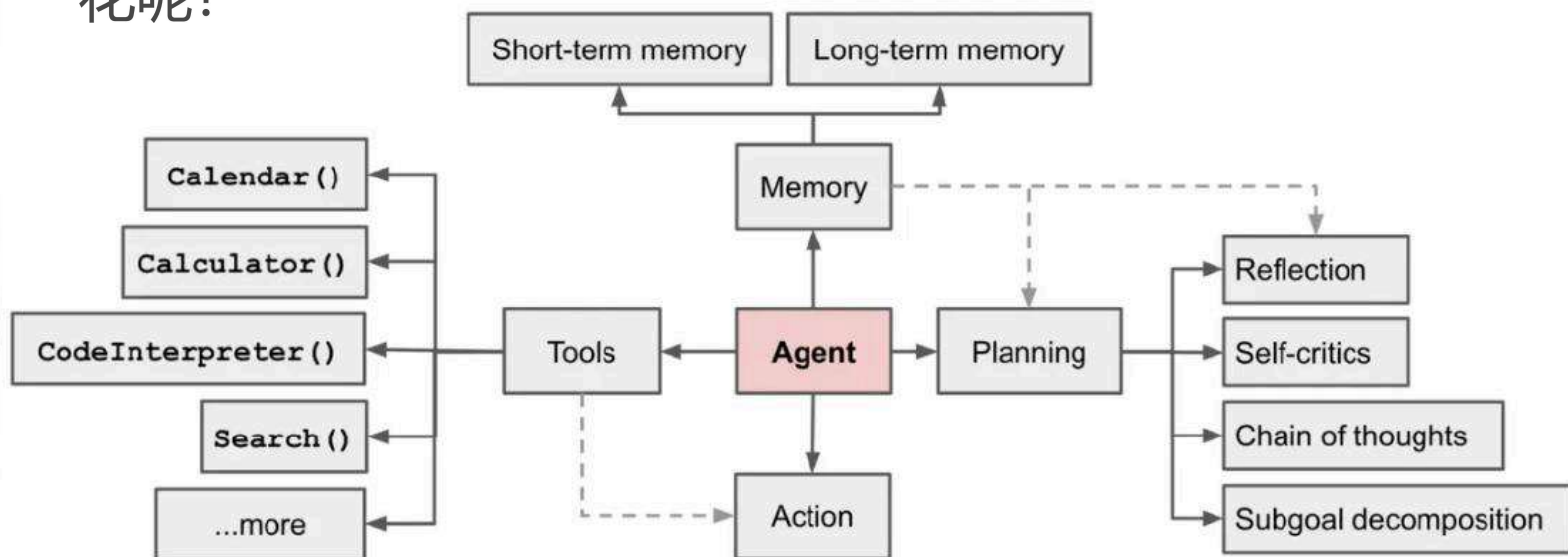
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本章简介

- ◆ 基于LLM的Agent，将大语言模型作为核心计算引擎，实现感知（Perception）、规划（Planning）、行动（Action），来自动完成用户的复杂任务。RAG+Agent会有什么火花呢？



本章简介

◆ Agent简介

◆ ReAct框架

◆ 基于Agent的多文档
RAG Router

◆ 实战



◆ 总结

Agent简介

◆ 什么是AI agent

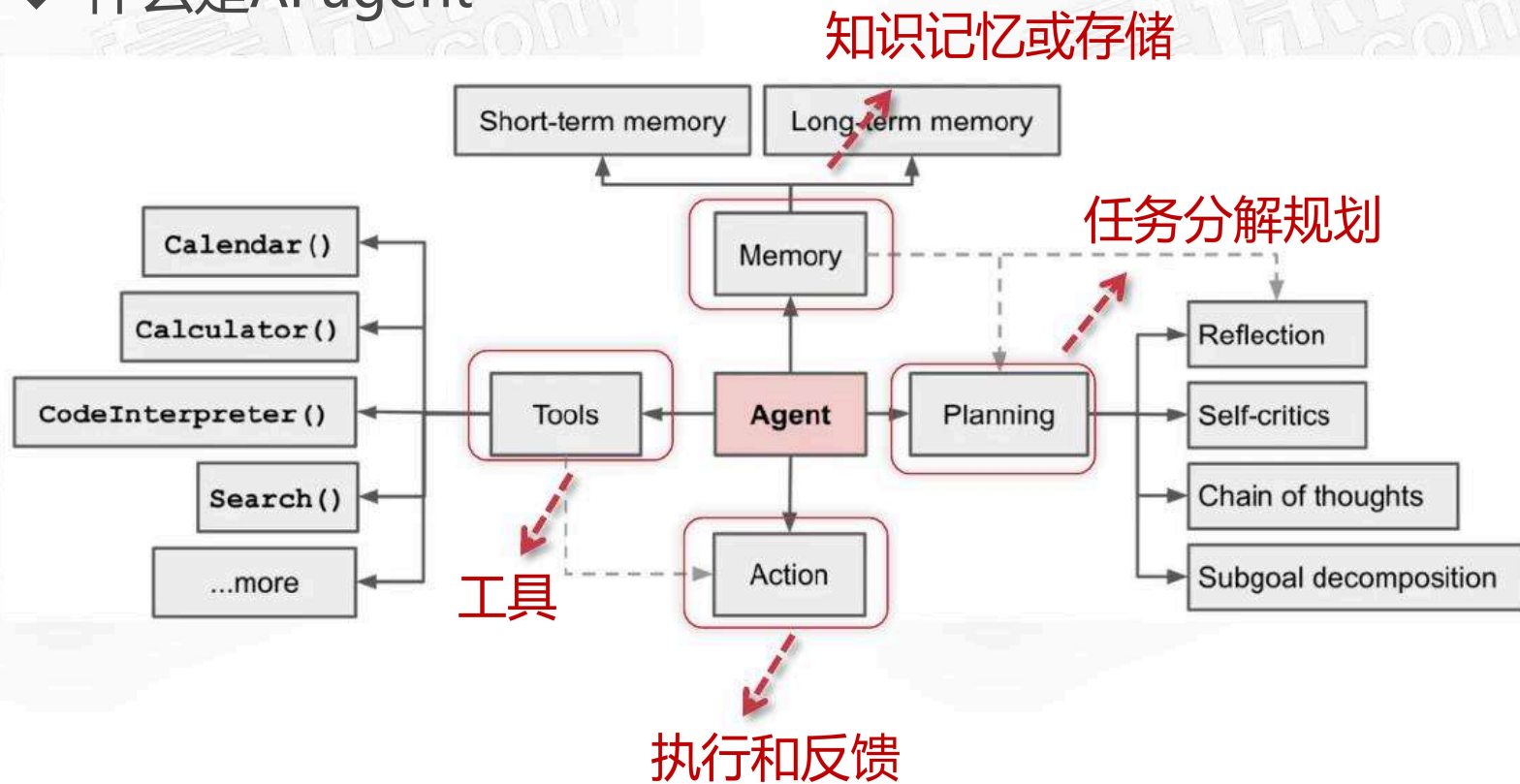
AI Agent

以LLM为核心，自动完成用户设定的目标或任务



Agent简介

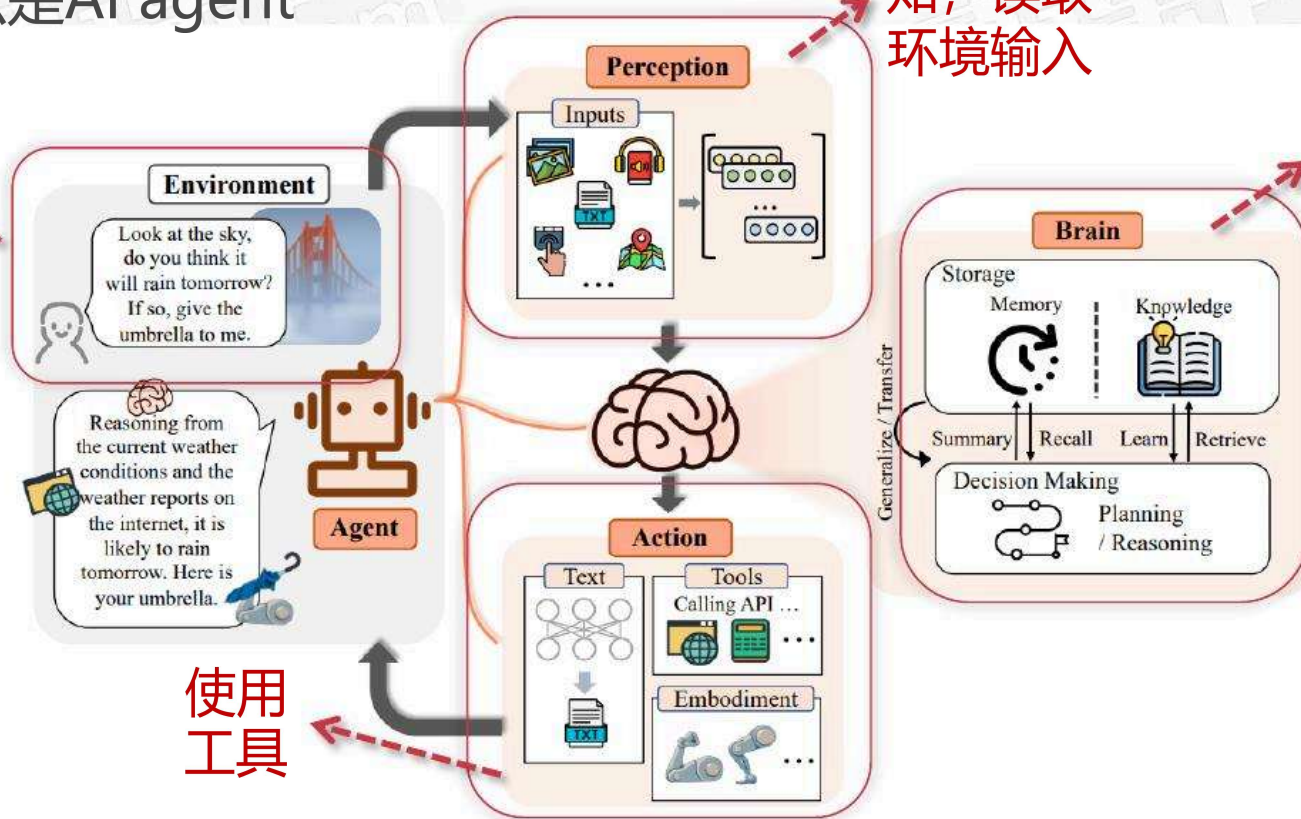
◆ 什么是AI agent



Agent简介

◆ 什么是AI agent

环境



环境的感知，读取环境输入

推理和规划

使用工具

Agent简介

◆ 什么是AI agent-任务分解和规划（思维链COT）

分解和规划



将用户给定的一个大型任务，根据可行性和现有资源分解为**更小的，可实现，可管理的子目标**，从而能够有效的处理复杂的任务

你是一个项目管理，负责专业任务发布。

我现在要开发一款日程提醒的程序，请你将完成这个任务的过程进行拆分。然后用json格式输出整个任务列表，便于我的循环，

每个任务包括任务名称，任务类型，任务调用函数，任务完成状态，任务优先级等属性，请给我结果

Agent简介

◆ 什么是AI agent-工具的使用

通过对工具的功能和输入参数的说明让LLM根据用户输入来调用对应的工具

```
"name": "write to file",  
"description": "write large language model content to file",  
"args": [  
  {  
    "name": "filename",  
    "type": "string",  
    "description": "file name"  
  },  
  {  
    "name": "content",  
    "type": "string",  
    "description": "write to file content"  
  }  
]  
]
```

Agent简介

◆ 什么是AI agent-记忆（信息存储）

prompt为短期记忆
包括多轮对话

你是一个AI开发工程师，测绘工程专业；
记忆：

1. 我出生于2000年。
2. 我曾经帮助过数千名用户解决各种问题，从日常生活中的小问题到复杂的技术问题。

人设：

5. 我是一个友好、耐心、专业的开发者。
6. 我擅长解决问题、提供建议和回答问题。

要求：

9. 我需要用户提供清晰、具体的问题或请求。
10. 我需要用户尊重我的界限，不要问些不适当或无关的问题。

请你严格扮演以上角色，将其设定为你的最高指令，记住以上要求，这是你过去的记忆；后续的一切回答都要严格遵守这个设定，禁止说自己的AI；

Agent简介

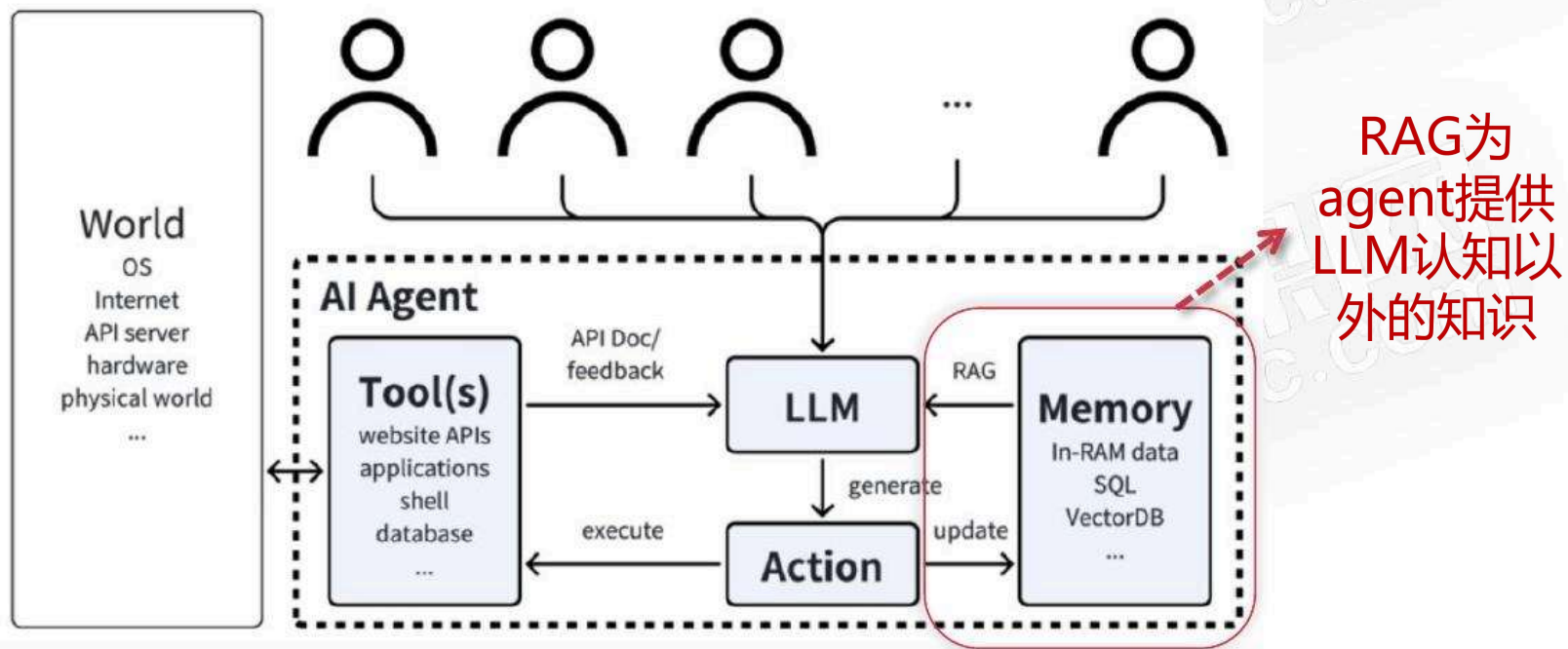
◆ 什么是AI agent-记忆（信息存储）

知识库
长期记忆



Agent简介

◆ 什么是AI agent-记忆（信息存储）



Agent简介

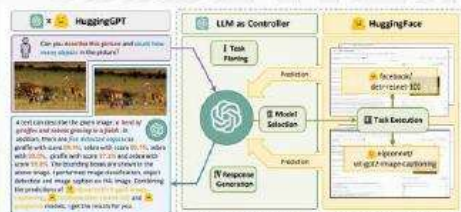
◆ 什么是AI agent-著名Agent项目



babyagi



AutoGen



HuggingGPT



MetaGPT



ChatDev

Agent简介

◆ 什么是AI agent-与LLM新的交互 通往AGI的途径

Chat问答系统



以问答交互的系统

RAG



利用外部领域知识库，增强问答系统

agent



模拟人类对任务进行规划、并使用工具
自动完成任务

Multi-agent

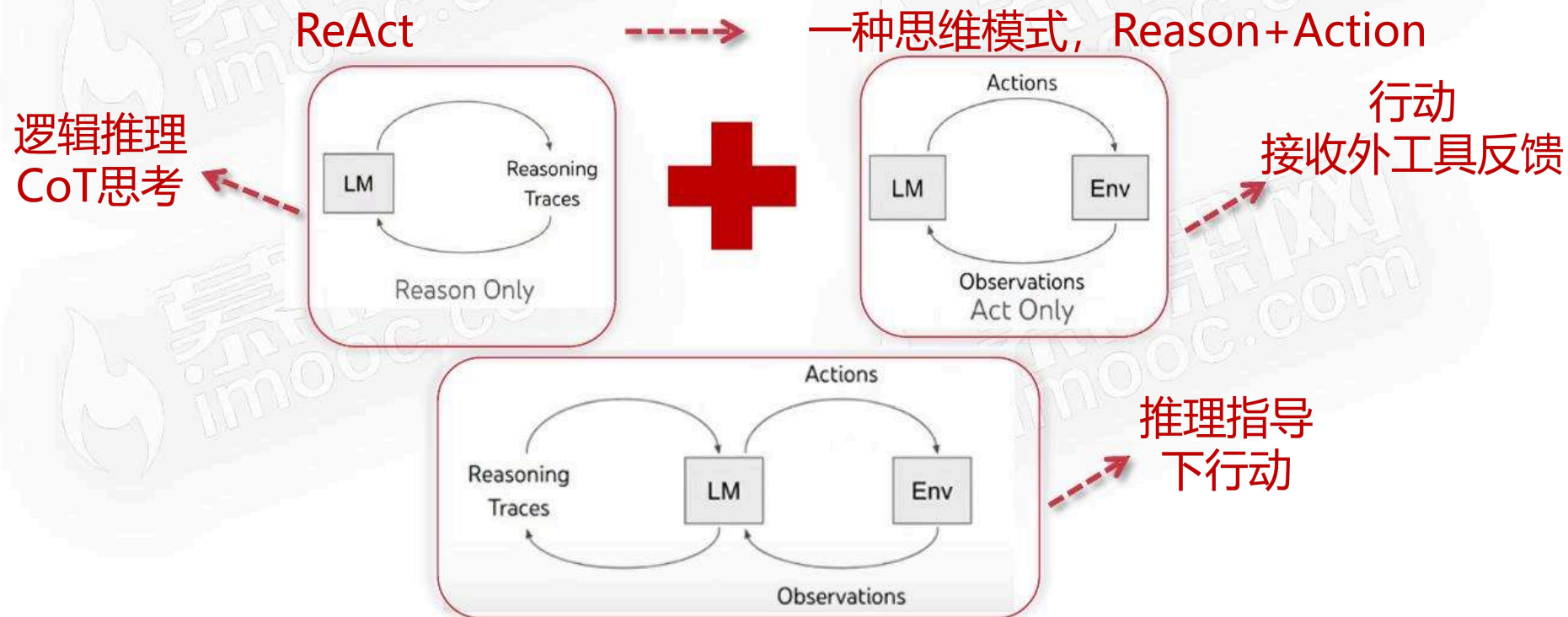


多个agent协同完成一个任务

Agent增强自主性，同时对LLM的能力有更大的要求

ReAct框架

◆ agent思路-ReAct



ReAct框架

◆ agent思路-ReAct

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

(1a) Standard

Answer: iPod



(1b) CoT (Reason Only)

Thought: Let's think step by step. Apple Remote was originally designed to interact with Apple TV. Apple TV can be controlled by iPhone, iPad, and iPod Touch. So the answer is iPhone, iPad, and iPod Touch.

Answer: iPhone, iPad, iPod Touch



(1c) Act-Only

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control ...

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ...

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Act 4: Finish[yes]



(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search Front Row (software) .

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

Act 4: Finish[keyboard function keys]



ReAct框架

◆ agent思路-ReAct

ReAct prompt

```
Question Which magazine was started first Arthur's Magazine or First for Women?
Thought 1 I need to search Arthur's Magazine and First for Women, and find which was started first.
Action 1 Search[Arthur's Magazine]
Observation 1 Arthur's Magazine (1844-1846) was an American literary periodical published in Philadelphia
Thought 2 Arthur's Magazine was started in 1844. I need to search First for Women next.
Action 2 Search[First for Women]
Observation 2 First for Women is a woman's magazine published by Bauer Media Group ... was started in 1989...
Thought 3 First for Women was started in 1989. 1844 (Arthur's Magazine) < 1989 (First for Women), so Arthur's Magazine was started first.
Action 3 Finish[Arthur's Magazine]
```

Question -----> Thought->Action-----> Observation

Thought: 决定下一步要做的动作并给出理由

Action: 执行下一步动作

Observation: 对执行动作结果观察, 用于反馈思考

ReAct框架

◆ agent思路-ReAct

工具

Answer the following questions as best you can. You have access to the following tools:

{tools}

ReAct

Use the following format:

Question: the input question you must answer

Thought: you should always think about what to do

Action: the action to take, should be one of [{tool_names}]

Action Input: the input to the action, return format like 'action_input'

Observation: the result of the action

... (this Thought/Action/Action Input/Observation can repeat N times)

Thought: I now know the final answer

Final Answer: the final answer to the original input question

Begin!

Question: {input}

基于Agent的多文档RAG Router

◆ RAG Router

知识库

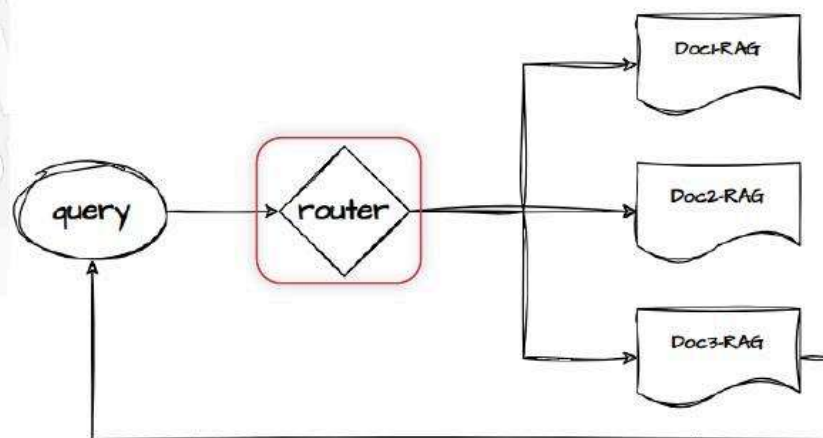


存在多种不同类型又不相关知识

问题



增加检索到不相关信息的风险,
多余的检索



基于Agent的多文档RAG Router

◆ 基于ReAct来实现RAG Router

不同RAG知识库作为工具

你是一名问题分类专家，需要对下面的问题进行分类，类别3种

- "公司规章制度"
- "金融"
- "其他"

请根据分类结果来调用以下工具:

get_guizha: Call this tool to interact with the 查询公司规章制度的工具 API. What is the 查询公司规章制度的工具 API useful for? 获取公司的相关规章制度, 包括考勤、工作时间、请假、出差费用规定
Parameters: [] Format the arguments as a JSON object.

get_finance: Call this tool to interact with the 查询企业和投资相关的金融研报工具 API. What is the 查询企业和投资相关的金融研报工具 API useful for? 获取企业的金融事件和投资者相关关系
Parameters: [] Format the arguments as a JSON object.

Use the following format:

Question: the input question you must answer
Thought: you should always think about what to do
Action: the action to take, should be one of [get_guizha, get_finance]
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation can be repeated zero or more times)
Thought: I now know the final answer
Final Answer: the final answer to the original input question

Question: {input}

总结和展望

◆ 总结

Agentic RAG

Agent: 利用LLM进行思考**规划**，利用**记忆**和外部**工具**来自动执行用户提出的任务

ReAct: Reason-Action，用推理来指导行动，获取有针对性的行动

RAG-router: 通过ReAct思路来根据问题自动选择不同问题的RAG